

Reviewer Pack — Minimal Order of Local Systems over Curves vs DPLL Proof Tim...

Entry 08eb958cb85e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-24 23:19:27 UTC

Minimal Order of Local Systems over Curves vs DPLL Proof Time Complexity

Entry ID: 08eb958cb85e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 23:19:27 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Local systems over curves) **Field B** (complexity object): Complexity Theory: DPLL Proof Time Complexity

Statement:

[‘For every n -variable 3-CNF formula F , the minimal order of a local system on a curve X that classifies the automorphism group of its affine scheme $\text{Spec}(\mathbb{Z}[X]/(F))$ is $\Theta(t(F)^{1/2})$, where $t(F)$ is the minimum DPLL proof time for F .’, ‘Equivalently, the minimal order of such a local system is lower-bounded by $\Omega(\log_2 t^*(F))$.’]

Rationale (proposer’s reasoning):

[‘Local systems over curves provide a rich algebraic-geometric framework to study symmetry in algebraic structures, which can be related to computational complexity through the DPLL proof time. The conjecture aims to bridge these two fields by proposing a quantitative relationship between the geometric invariant and the computational complexity of solving 3-CNF formulas.’]

Taxonomy category: local_systems_curve_algebraic_geometry (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ee39b3ea09a94b60

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture will be supported if, for at least 80% of the n -variable 3-CNF formulas with $n \in \{12, 14, 16, 18, 20\}$, the minimal order of a local system on a constructed curve X is within $\pm 30\%$ of $\Theta(t(F)^{1/2})$, where $t(F)$ is the minimum DPLL proof time for F . The conjecture will be falsified if any seed produces a minimal order that deviates by more than 30% from $\Theta(t^*(F)^{1/2})$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - algebraic geometry AND local systems over curves AND DPLL proof time complexity - DPLL proof time complexity IN algebraic geometry AND local systems - local system order $\Theta(t^*(F)^{(1/2)})$ AND $\Omega(\log_2 t^*(F))$

Top relevant hits considered: - [http://arxiv.org/abs/1912.00347v3] Equiresidual algebraic geometry I: The affine theory - [http://arxiv.org/abs/1311.2051v5] Some results from algebraic geometry over complete discretely valued fields - [http://arxiv.org/abs/1312.2935v6] Some results from algebraic geometry over Henselian real valued fields - [http://arxiv.org/abs/0803.1825v1] On the algebraic geometry of polynomial dynamical systems - [http://arxiv.org/abs/1607.00443v1] Algebraic Proof Complexity: Progress, Frontiers and Challenges - [http://arxiv.org/abs/gr-qc/9810059v1] Space-time distributions - [http://arxiv.org/abs/0901.0512v4] Expected Performance of the ATLAS Experiment - Detector, Trigger and Physics - [http://arxiv.org/abs/2603.25938v1] Narrowband searches for continuous gravitational waves from known pulsars in the first two parts of the fourth LIGO-Vir

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(-n, -1), random.randint(1, n)]
            random.shuffle(clause)
            clauses.append(clause)
        return clauses

    def dpll_time_complexity(clauses):
        # Simplified DPLL complexity estimation
        return 2 ** len(clauses)

    def local_system_order(t_star):
```

```

    # Simplified order estimation
    return int(math.log2(t_star))

n = random.choice([12, 14, 16, 18, 20])
formula = generate_3cnf(n)
t_star = dpll_time_complexity(formula)
order = local_system_order(t_star)

expected_order = math.isclose(order, t_star ** 0.5, rel_tol=0.3) or math.isclose(order, t_star

return {
    "metric_name": "local_system_order",
    "metric_value": order,
    "instances_tested": 1,
    "conjecture_holds": expected_order,
    "counterexample": f"n={n}, t_star={t_star}, order={order}, expected_order={expected_order}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(s for s, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\n n={results[0]['instances_tested']}, t_star={results[first_failing_seed]['t_star']}")
    else:
        print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

56, order=28, expected_order=False}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 24, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=28, order=24, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 32, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=32, order=32, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 32, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=32, order=32, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 24, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=24, order=24, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=40, order=40, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 28, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=28, order=28, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 40, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=40, order=40, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 28, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=28, order=28, expected_order=False'}
TRIAL: {'metric_name': 'local_system_order', 'metric_value': 28, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=56, t_star=28, order=28, expected_order=False'}

```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_34a13f57.py", line 70, in <module>
    print(f"RESULT: FALSIFIED counterexample=\n={results[0]['instances_tested']}, t_star={re
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test produced counterexamples where the minimal order of a local system deviates significantly from the expected order based on DPLL proof time co | next: Investigate the cause of the KeyError and address it to ensure accurate testing. Additionally, explore alternative methods or refine the conjecture to account for the observed deviations.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11898
2	preregistration	ollama_remote	glm4:latest	0	0	6758
3	novelty	ollama_remote	glm4:latest	0	0	4750
4	novelty	ollama_remote	glm4:latest	0	0	6387
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19568
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10479
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9030
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8654
9	judge	ollama_remote	glm4:latest	0	0	10553

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 88076 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/08eb958cb85e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/08eb958cb85e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/08eb958cb85e.tar.gz (if generated)